

B5 General Relativity, 2025 — Model Solutions

Question 1 — Covariant box of a scalar field

Problem. For a scalar $f(x^\mu)$ on $(g_{\alpha\beta}, x^\mu)$: (a) show $\nabla_\mu \nabla_\nu f = \partial_\mu \partial_\nu f - \Gamma_{\mu\nu}^\lambda \partial_\lambda f$; (b) using $\partial_\lambda g = g g^{\mu\nu} \partial_\lambda g_{\mu\nu}$, prove $\Gamma_{\mu\lambda}^\mu = (\sqrt{-g})^{-1} \partial_\lambda \sqrt{-g}$; (c) deduce $g^{\mu\nu} \nabla_\mu \nabla_\nu f = (\sqrt{-g})^{-1} \partial_\alpha (g^{\alpha\beta} \sqrt{-g} \partial_\beta f)$; (d) for $U_{\mu\nu} = \nabla_\mu \nabla_\nu f - g_{\mu\nu} \nabla^\lambda \nabla_\lambda f$, show $\nabla^\mu U_{\mu\nu} = R_\nu{}^\sigma \nabla_\sigma f$ and $\nabla^\nu \nabla^\mu U_{\mu\nu} = \frac{1}{2} \nabla^\nu R \nabla_\nu f + R^{\nu\mu} \nabla_\nu \nabla_\mu f$; (e) in vacuum with Λ , $\nabla^\mu \nabla^\nu U_{\mu\nu} = \frac{1}{3} \Lambda g^{\mu\nu} U_{\mu\nu}$.

(a)

Scalar covariant derivative reduces to partial: $\nabla_\nu f = \partial_\nu f$. The one-form $\nabla_\nu f$ then carries one lower index:

$$\nabla_\mu \nabla_\nu f = \partial_\mu (\nabla_\nu f) - \Gamma_{\mu\nu}^\lambda \nabla_\lambda f.$$

Insert $\nabla_\nu f = \partial_\nu f$:

$$\boxed{\nabla_\mu \nabla_\nu f = \partial_\mu \partial_\nu f - \Gamma_{\mu\nu}^\lambda \partial_\lambda f.} \quad (1)$$

Symmetric in (μ, ν) since $\Gamma_{\mu\nu}^\lambda = \Gamma_{\nu\mu}^\lambda$ (torsion-free).

(b)

Christoffel contraction:

$$\Gamma_{\mu\lambda}^\mu = \frac{1}{2} g^{\mu\nu} (\partial_\mu g_{\nu\lambda} + \partial_\lambda g_{\mu\nu} - \partial_\nu g_{\mu\lambda}).$$

First and third terms are equal under $\mu \leftrightarrow \nu$ via $g^{\mu\nu} = g^{\nu\mu}$, so they cancel:

$$\Gamma_{\mu\lambda}^\mu = \frac{1}{2} g^{\mu\nu} \partial_\lambda g_{\mu\nu}.$$

Apply Jacobi $\partial_\lambda g = g g^{\mu\nu} \partial_\lambda g_{\mu\nu}$:

$$\Gamma_{\mu\lambda}^\mu = \frac{1}{2g} \partial_\lambda g.$$

Rewrite as a logarithmic derivative (with $-g > 0$ for Lorentzian signature):

$$\Gamma_{\mu\lambda}^\mu = \frac{1}{2} \partial_\lambda \ln |g| = \frac{1}{2} \partial_\lambda \ln(-g).$$

Use $\ln(-g) = 2 \ln \sqrt{-g}$:

$$\Gamma_{\mu\lambda}^\mu = \partial_\lambda \ln \sqrt{-g}.$$

Expand the logarithmic derivative:

$$\Gamma_{\mu\lambda}^\mu = \frac{1}{\sqrt{-g}} \partial_\lambda \sqrt{-g}.$$

$$\boxed{\Gamma_{\mu\lambda}^\mu = \frac{1}{\sqrt{-g}} \partial_\lambda \sqrt{-g}.} \quad (2)$$

(c)

Box on a scalar:

$$g^{\mu\nu}\nabla_\mu\nabla_\nu f = \nabla_\mu(g^{\mu\nu}\partial_\nu f).$$

The argument $V^\mu \equiv g^{\mu\nu}\partial_\nu f$ is a vector; its divergence:

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma_{\mu\lambda}^\mu V^\lambda.$$

Insert (??):

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \frac{1}{\sqrt{-g}}(\partial_\lambda \sqrt{-g})V^\lambda.$$

Combine into a single product rule:

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}V^\mu).$$

Substitute $V^\mu = g^{\mu\nu}\partial_\nu f$:

$$\boxed{g^{\mu\nu}\nabla_\mu\nabla_\nu f = \frac{1}{\sqrt{-g}}\partial_\alpha(g^{\alpha\beta}\sqrt{-g}\partial_\beta f).} \quad (3)$$

(d)

Definition $U_{\mu\nu} = \nabla_\mu\nabla_\nu f - g_{\mu\nu}\square f$ with $\square f \equiv \nabla^\lambda\nabla_\lambda f$. Take divergence using $\nabla_\mu g_{\alpha\beta} = 0$:

$$\nabla^\mu U_{\mu\nu} = \nabla^\mu\nabla_\mu\nabla_\nu f - \nabla_\nu\square f.$$

Use symmetry $\nabla_\mu\nabla_\nu f = \nabla_\nu\nabla_\mu f$ from (??):

$$\nabla^\mu\nabla_\mu\nabla_\nu f = \nabla^\mu\nabla_\nu\nabla_\mu f.$$

Insert the commutator $[\nabla^\mu, \nabla_\nu]\nabla_\mu f$:

$$\nabla^\mu\nabla_\nu\nabla_\mu f = \nabla_\nu\nabla^\mu\nabla_\mu f + [\nabla^\mu, \nabla_\nu]\nabla_\mu f.$$

Apply the Riemann identity on the one-form $\omega_\alpha = \nabla_\alpha f$ (sign flip from vector form):

$$[\nabla_\mu, \nabla_\nu]\nabla_\alpha f = -R^\beta_{\alpha\mu\nu}\nabla_\beta f.$$

Raise the first index with $g^{\mu\sigma}$ and contract $\alpha \rightarrow \mu$:

$$[\nabla^\mu, \nabla_\nu]\nabla_\mu f = -g^{\mu\sigma}R^\beta_{\mu\sigma\nu}\nabla_\beta f.$$

Use antisymmetry of Riemann on last two indices, $R^\beta_{\mu\sigma\nu} = -R^\beta_{\mu\nu\sigma}$:

$$g^{\mu\sigma}R^\beta_{\mu\sigma\nu} = -g^{\mu\sigma}R^\beta_{\mu\nu\sigma}.$$

Raise μ on the right:

$$g^{\mu\sigma}R^\beta_{\mu\sigma\nu} = -R^{\beta\sigma}_{\nu\sigma}.$$

Swap the two lower indices (Riemann antisymmetry), $R^{\beta\sigma}_{\nu\sigma} = -R^{\beta\sigma}_{\sigma\nu}$:

$$g^{\mu\sigma}R^\beta_{\mu\sigma\nu} = R^{\beta\sigma}_{\sigma\nu}.$$

Apply first-pair antisymmetry of Riemann, $R^{\beta\sigma}{}_{\sigma\nu} = -R^{\sigma\beta}{}_{\sigma\nu}$:

$$g^{\mu\sigma} R^{\beta}{}_{\mu\sigma\nu} = -R^{\sigma\beta}{}_{\sigma\nu}.$$

Apply paper's Ricci convention $R_{\alpha\nu} \equiv -R^{\sigma}{}_{\alpha\nu\sigma} = R^{\sigma}{}_{\alpha\sigma\nu}$:

$$R^{\sigma\beta}{}_{\sigma\nu} = R^{\beta}{}_{\nu}.$$

Combine:

$$g^{\mu\sigma} R^{\beta}{}_{\mu\sigma\nu} = -R^{\beta}{}_{\nu}.$$

Substitute back:

$$[\nabla^{\mu}, \nabla_{\nu}] \nabla_{\mu} f = R^{\beta}{}_{\nu} \nabla_{\beta} f.$$

Substitute into $\nabla^{\mu} U_{\mu\nu}$:

$$\nabla^{\mu} U_{\mu\nu} = \nabla_{\nu} \square f + R^{\beta}{}_{\nu} \nabla_{\beta} f - \nabla_{\nu} \square f.$$

Cancel $\nabla_{\nu} \square f$:

$$\boxed{\nabla^{\mu} U_{\mu\nu} = R_{\nu}{}^{\sigma} \nabla_{\sigma} f.} \quad (4)$$

Take a second divergence:

$$\nabla^{\nu} \nabla^{\mu} U_{\mu\nu} = \nabla^{\nu} (R_{\nu}{}^{\sigma} \nabla_{\sigma} f).$$

Apply the product rule:

$$\nabla^{\nu} \nabla^{\mu} U_{\mu\nu} = (\nabla^{\nu} R_{\nu}{}^{\sigma}) \nabla_{\sigma} f + R_{\nu}{}^{\sigma} \nabla^{\nu} \nabla_{\sigma} f.$$

Start from the Bianchi identity:

$$\nabla_{\gamma} R^{\mu}{}_{\nu\alpha\beta} + \nabla_{\beta} R^{\mu}{}_{\nu\gamma\alpha} + \nabla_{\alpha} R^{\mu}{}_{\nu\beta\gamma} = 0.$$

Contract first by setting $\beta = \mu$ (sum):

$$\nabla_{\gamma} R^{\mu}{}_{\nu\alpha\mu} + \nabla_{\mu} R^{\mu}{}_{\nu\gamma\alpha} + \nabla_{\alpha} R^{\mu}{}_{\nu\mu\gamma} = 0.$$

Apply paper's Ricci convention $R_{\nu\alpha} = -R^{\mu}{}_{\nu\alpha\mu}$ and antisymmetry of last two indices, $R^{\mu}{}_{\nu\mu\gamma} = -R^{\mu}{}_{\nu\gamma\mu} = R_{\nu\gamma}$:

$$-\nabla_{\gamma} R_{\nu\alpha} + \nabla_{\mu} R^{\mu}{}_{\nu\gamma\alpha} + \nabla_{\alpha} R_{\nu\gamma} = 0.$$

Contract with $g^{\nu\alpha}$ to bring out the Ricci scalar:

$$-\nabla_{\gamma} R + \nabla^{\mu} R_{\mu\gamma} + \nabla^{\mu} R_{\mu\gamma} = 0.$$

Combine the divergence terms:

$$\nabla_{\gamma} R - 2\nabla^{\mu} R_{\mu\gamma} = 0.$$

Solve for the divergence of Ricci:

$$\nabla^{\nu} R_{\nu}{}^{\sigma} = \frac{1}{2} \nabla^{\sigma} R.$$

Insert into the second divergence:

$$\boxed{\nabla^{\nu} \nabla^{\mu} U_{\mu\nu} = \frac{1}{2} \nabla^{\sigma} R \nabla_{\sigma} f + R^{\nu\mu} \nabla_{\nu} \nabla_{\mu} f.} \quad (5)$$

(e)

Vacuum Einstein with Λ and $T_{\alpha\beta} = 0$:

$$R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R - \Lambda g_{\alpha\beta} = 0.$$

Trace with $g^{\alpha\beta}$ in 4D:

$$R - 2R - 4\Lambda = 0.$$

Solve for R :

$$R = -4\Lambda.$$

Substitute $R = -4\Lambda$ back into the field equation:

$$R_{\alpha\beta} = \frac{1}{2}g_{\alpha\beta}(-4\Lambda) + \Lambda g_{\alpha\beta}.$$

Simplify:

$$R_{\alpha\beta} = -\Lambda g_{\alpha\beta}.$$

Constant R gives $\nabla^\sigma R = 0$; the first term of (??) drops:

$$\nabla^\nu \nabla^\mu U_{\mu\nu} = R^{\nu\mu} \nabla_\nu \nabla_\mu f.$$

Insert $R^{\nu\mu} = -\Lambda g^{\nu\mu}$:

$$\nabla^\nu \nabla^\mu U_{\mu\nu} = -\Lambda g^{\nu\mu} \nabla_\nu \nabla_\mu f = -\Lambda \square f.$$

Trace $U_{\mu\nu}$ with $g^{\mu\nu}$ in 4D:

$$g^{\mu\nu} U_{\mu\nu} = \square f - 4\square f.$$

Combine:

$$g^{\mu\nu} U_{\mu\nu} = -3\square f.$$

Solve for $\square f$:

$$\square f = -\frac{1}{3}g^{\mu\nu} U_{\mu\nu}.$$

Substitute into $\nabla^\nu \nabla^\mu U_{\mu\nu} = -\Lambda \square f$:

$$\nabla^\mu \nabla^\nu U_{\mu\nu} = \frac{1}{3}\Lambda g^{\mu\nu} U_{\mu\nu}.$$

 (6)

Question 2 — Schwarzschild geodesics and the angular size of a star

Problem. Schwarzschild metric, $r_S = 2GM/c^2$. (a) Geodesic equations for a massive particle; gravitational redshift between r_{em} and r_{ob} ; behaviour at $r_{\text{em}} = r_S$. (b) Proper time from horizon to singularity for an object released from rest at $r = r_S$. (c) Two equatorial photons from r_0 : photon A aimed at $r = 0$, photon B grazing the surface at $r = R$ where $\dot{r} = 0$; show stated values of $(\dot{r}, \dot{\phi})$ at r_0 . (d) Angular diameter of a sphere of radius R as seen from r_0 . (e) Numerical angular size of Sgr A* using the light radius $r_L = \frac{3}{2}r_S$. (f) Why using r_S in place of r_L makes the formula diverge.

(a) Geodesics and gravitational redshift

Lagrangian (per unit mass, $\dot{} \equiv d/d\tau$):

$$2\mathcal{L} = -\left(1 - \frac{r_S}{r}\right) c^2 \dot{t}^2 + \left(1 - \frac{r_S}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2.$$

t and ϕ are cyclic. Conserved quantities:

$$E/m \equiv \left(1 - \frac{r_S}{r}\right) c^2 \dot{t} = \text{const}, \quad (7)$$

$$L/m \equiv r^2 \sin^2 \theta \dot{\phi} = \text{const}. \quad (8)$$

The θ -equation admits $\theta = \pi/2$; confine to the equator. Normalisation $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -c^2$:

$$-\left(1 - \frac{r_S}{r}\right) c^2 \dot{t}^2 + \left(1 - \frac{r_S}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = -c^2. \quad (9)$$

Solve (7) for $\dot{t}$ and (8) for $\dot{\phi}$:

$$\dot{t} = \frac{E}{mc^2(1 - r_S/r)}, \quad \dot{\phi} = \frac{L}{mr^2}.$$

Substitute into (9):

$$-\frac{E^2/(m^2 c^2)}{1 - r_S/r} + \frac{\dot{r}^2}{1 - r_S/r} + \frac{L^2}{m^2 r^2} = -c^2.$$

Multiply through by $(1 - r_S/r)$:

$$-\frac{E^2}{m^2 c^2} + \dot{r}^2 + \left(1 - \frac{r_S}{r}\right) \frac{L^2}{m^2 r^2} = -c^2 \left(1 - \frac{r_S}{r}\right).$$

Solve for $\dot{r}^2$:

$$\dot{r}^2 = \left(\frac{E}{mc}\right)^2 - \left(1 - \frac{r_S}{r}\right) \left(c^2 + \frac{L^2}{m^2 r^2}\right).$$

Redshift derivation. For a static emitter/observer $dr = d\theta = d\phi = 0$, so $ds^2 = -c^2 d\tau^2$ gives $d\tau = \sqrt{1 - r_S/r} dt$. A photon's frequency satisfies $\nu \propto 1/d\tau$, and the coordinate time dt between successive wave crests is common at r_{em} and r_{ob} (static metric):

$$\frac{\nu_{\text{ob}}}{\nu_{\text{em}}} = \frac{d\tau_{\text{em}}}{d\tau_{\text{ob}}}.$$

Substitute $d\tau = \sqrt{1 - r_S/r} dt$ at each point:

$$\boxed{\frac{\nu_{\text{ob}}}{\nu_{\text{em}}} = \sqrt{\frac{1 - r_S/r_{\text{em}}}{1 - r_S/r_{\text{ob}}}}}. \quad (10)$$

In the limit $r_{\text{em}} \rightarrow r_S$: $\nu_{\text{ob}}/\nu_{\text{em}} \rightarrow 0$, infinite redshift — the signal is frozen at the horizon.

(b) Proper time to the singularity

Object released from rest at $r = r_S$: $L = 0$ and the energy is fixed by $\dot{r} = 0$ at $r = r_S$. Radial geodesic energy equation from (a) with $L = 0$:

$$\dot{r}^2 = \left(\frac{E}{mc}\right)^2 - c^2 \left(1 - \frac{r_S}{r}\right).$$

Apply $\dot{r} = 0$ at $r = r_S$:

$$\left(\frac{E}{mc}\right)^2 = 0.$$

Substitute back into the energy equation:

$$\dot{r}^2 = -c^2\left(1 - \frac{r_S}{r}\right).$$

Rearrange (valid for $r < r_S$):

$$\dot{r}^2 = c^2\left(\frac{r_S}{r} - 1\right).$$

Take the negative (infalling) root:

$$\frac{dr}{d\tau} = -c\sqrt{\frac{r_S - r}{r}}.$$

Separate variables:

$$d\tau = -\frac{1}{c}\sqrt{\frac{r}{r_S - r}} dr.$$

Substitute $x = r/r_S$, $dr = r_S dx$:

$$d\tau = -\frac{r_S}{c}\sqrt{\frac{x}{1-x}} dx.$$

Integrate from $r = r_S$ ($x = 1$) to $r = 0$ ($x = 0$):

$$\Delta\tau = \frac{r_S}{c} \int_0^1 \sqrt{\frac{x}{1-x}} dx.$$

Apply hint $\int_0^1 \sqrt{x/(1-x)} dx = \pi/2$:

$$\boxed{\Delta\tau = \frac{\pi r_S}{2c}.} \quad (11)$$

(c) Photon trajectories from r_0

Photon: $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$. Equator, conserved energy and angular momentum:

$$\mathcal{E} = \left(1 - \frac{r_S}{r}\right) c\dot{t}, \quad \mathcal{L} = r^2\dot{\phi}.$$

Null condition:

$$\left(1 - \frac{r_S}{r}\right) c^2\dot{t}^2 = \left(1 - \frac{r_S}{r}\right)^{-1} \dot{r}^2 + r^2\dot{\phi}^2.$$

Substitute $\dot{t}, \dot{\phi}$:

$$\dot{r}^2 = \mathcal{E}^2 - \left(1 - \frac{r_S}{r}\right) \frac{\mathcal{L}^2}{r^2}. \quad (12)$$

Photon A. Aimed radially at $r = 0$: $\mathcal{L} = 0$ and $\dot{\phi} = 0$. Then (??) gives $\dot{r}^2 = \mathcal{E}^2$, so

$$\dot{r} = -\mathcal{E} = -ck, \quad k \equiv \mathcal{E}/c.$$

$$\boxed{(\dot{r}, \dot{\phi})_A = ck(-1, 0).} \quad (13)$$

Photon B. Turning point at $r = R$ gives $\dot{r}(R) = 0$ in (??):

$$\mathcal{E}^2 = \left(1 - \frac{r_S}{R}\right) \frac{\mathcal{L}^2}{R^2}.$$

Solve for $\mathcal{L}$:

$$\mathcal{L} = \mathcal{E} R \left(1 - \frac{r_S}{R}\right)^{-1/2}.$$

Evaluate $\dot{\phi} = \mathcal{L}/r_0^2$ at $r = r_0$:

$$\dot{\phi} = \frac{\mathcal{E} R}{r_0^2} \left(1 - \frac{r_S}{R}\right)^{-1/2}.$$

Substitute $\mathcal{E} = ck$:

$$\dot{\phi} = ck \left(1 - \frac{r_S}{R}\right)^{-1/2} \frac{R}{r_0^2}.$$

Evaluate (??) at $r = r_0$ with $\mathcal{L} = \mathcal{E} R (1 - r_S/R)^{-1/2}$:

$$\dot{r}^2 = \mathcal{E}^2 - \left(1 - \frac{r_S}{r_0}\right) \frac{\mathcal{E}^2 R^2}{r_0^2 (1 - r_S/R)}.$$

Factor $\mathcal{E}^2 (1 - r_S/R)^{-1}$:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{1 - r_S/R} \left[\left(1 - \frac{r_S}{R}\right) - \frac{R^2}{r_0^2} \left(1 - \frac{r_S}{r_0}\right) \right].$$

Reorganise the bracket into the form requested by the question (the exact bracket equals the factored form only at zeroth order in r_S/r ; the difference $r_S(1/R - 1/r_0)$ is dropped):

$$\left(1 - \frac{r_S}{R}\right) - \frac{R^2}{r_0^2} \left(1 - \frac{r_S}{r_0}\right) \simeq \left(1 - \frac{R^2}{r_0^2}\right) \left(1 - \frac{r_S}{r_0}\right).$$

Insert:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{1 - r_S/R} \left(1 - \frac{R^2}{r_0^2}\right) \left(1 - \frac{r_S}{r_0}\right).$$

Take the inward root and substitute $\mathcal{E} = ck$:

$$\boxed{(\dot{r}, \dot{\phi})_B = ck \left(1 - \frac{r_S}{R}\right)^{-1/2} \left(- \left(1 - \frac{R^2}{r_0^2}\right)^{1/2} \left(1 - \frac{r_S}{r_0}\right)^{1/2}, \frac{R}{r_0^2} \right).} \quad (14)$$

(d) Angular size of the star

Static observer at r_0 uses orthonormal tetrad with $e_{\hat{t}} = (1 - r_S/r_0)^{-1/2} \partial_t$, $e_{\hat{r}} = (1 - r_S/r_0)^{1/2} \partial_r$, $e_{\hat{\phi}} = r_0^{-1} \partial_{\phi}$. Tetrad components of a 4-velocity $\dot{x}^\mu$:

$$\hat{v}^{\hat{t}} = \left(1 - \frac{r_S}{r_0}\right)^{1/2} c \dot{t}, \quad \hat{v}^{\hat{r}} = \left(1 - \frac{r_S}{r_0}\right)^{-1/2} \dot{r}, \quad \hat{v}^{\hat{\phi}} = r_0 \dot{\phi}.$$

Compute $c \dot{t}_B$ from conservation $\mathcal{E} = (1 - r_S/r_0) c \dot{t}$:

$$c \dot{t}_B = \frac{\mathcal{E}}{1 - r_S/r_0} = \frac{ck}{1 - r_S/r_0}.$$

Hence $\hat{v}_B^{\hat{t}}$ for photon B :

$$\hat{v}_B^{\hat{t}} = \left(1 - \frac{r_S}{r_0}\right)^{1/2} \frac{ck}{1 - r_S/r_0} = \frac{ck}{(1 - r_S/r_0)^{1/2}}.$$

For a null 4-velocity $\hat{v}^{\hat{t}} = |\hat{\mathbf{v}}|$; the spatial unit direction is $\hat{\mathbf{n}} = \hat{\mathbf{v}}/\hat{v}^{\hat{t}}$. Photon A is purely radial, so $\hat{\mathbf{n}}_A = (-1, 0)$. For photon B the tangential component:

$$\hat{n}_B^{\hat{\phi}} = \frac{\hat{v}_B^{\hat{\phi}}}{\hat{v}_B^{\hat{t}}} = \frac{r_0 \dot{\phi}_B}{ck/(1 - r_S/r_0)^{1/2}}.$$

Substitute $\dot{\phi}_B = ck(1 - r_S/R)^{-1/2}R/r_0^2$:

$$\hat{n}_B^{\hat{\phi}} = \frac{r_0 \cdot ck(1 - r_S/R)^{-1/2}R/r_0^2}{ck/(1 - r_S/r_0)^{1/2}}.$$

Simplify:

$$\hat{n}_B^{\hat{\phi}} = \frac{R(1 - r_S/r_0)^{1/2}}{r_0(1 - r_S/R)^{1/2}}.$$

The half angle from the radial direction satisfies $\sin(\phi/2) = \hat{n}_B^{\hat{\phi}}$:

$$\sin\left(\frac{\phi}{2}\right) = \frac{R}{r_0} \sqrt{\frac{1 - r_S/r_0}{1 - r_S/R}}.$$

$$\boxed{\phi = 2 \arcsin \left[\frac{R}{r_0} \sqrt{\frac{1 - r_S/r_0}{1 - r_S/R}} \right]}. \quad (15)$$

Newtonian limit $r_S \rightarrow 0$ gives $\phi \rightarrow 2 \arcsin(R/r_0) \approx 2R/r_0$, as required.

(e) Angular size of Sgr A*

Data: $M_1 = 4.3 \times 10^6 M_\odot$, $r_{S\odot} = 9.5 \times 10^{-17}$ kpc, $r_1 = 8$ kpc, $R = r_L = \frac{3}{2}r_S$.

Compute the Schwarzschild radius:

$$r_S = 4.3 \times 10^6 \cdot 9.5 \times 10^{-17} \text{ kpc}.$$

Multiply:

$$r_S = 4.085 \times 10^{-10} \text{ kpc}.$$

Compute the light radius:

$$r_L = 1.5 r_S = 6.13 \times 10^{-10} \text{ kpc}.$$

Evaluate the redshift factors with $R = r_L$, $r_0 = r_1 = 8$ kpc:

$$1 - \frac{r_S}{r_0} \approx 1.$$

$$1 - \frac{r_S}{r_L} = 1 - \frac{2}{3} = \frac{1}{3}.$$

Insert into (??):

$$\frac{R}{r_0} \sqrt{\frac{1 - r_S/r_0}{1 - r_S/R}} = \frac{6.13 \times 10^{-10}}{8} \sqrt{3}.$$

Evaluate the prefactor:

$$\frac{R}{r_0} = 7.66 \times 10^{-11}.$$

Multiply by $\sqrt{3} \approx 1.732$:

$$\sin(\phi/2) \approx 7.66 \times 10^{-11} \cdot 1.732 \approx 1.33 \times 10^{-10}.$$

Small-angle approximation $\phi \approx 2 \sin(\phi/2)$:

$$\phi \approx 2.66 \times 10^{-10} \text{ rad.}$$

Convert using $1 \text{ rad} = 2.063 \times 10^{11} \mu\text{as}$:

$$\phi \approx 2.66 \times 10^{-10} \cdot 2.063 \times 10^{11} \mu\text{as}.$$

Multiply:

$$\phi \approx 55 \mu\text{as}.$$

$$\boxed{\phi_{\text{Sgr A}^*} \approx 2.7 \times 10^{-10} \text{ rad} \approx 55 \mu\text{as}.} \quad (16)$$

Consistent with the EHT image ($\sim 50 \mu\text{as}$).

(f) Why $R = r_S$ would diverge

Set $R = r_S$ in (??):

$$1 - \frac{r_S}{R} = 0.$$

The argument of arcsin scales as $(1 - r_S/R)^{-1/2} \rightarrow \infty$. Physically: a photon emitted tangentially exactly at $r = r_S$ is on the horizon and cannot escape; the locally measured tangential velocity must equal c , forcing the apparent angular size to fill the sky. The light radius $r_L = \frac{3}{2}r_S$ is the smallest closed-orbit radius for null geodesics (photon sphere); any object with $R < r_L$ still has an apparent edge at r_L from far away.

Question 3 — Wormhole geometry

Problem. Line element $ds^2 = -c^2 dt^2 + dr^2 + (R^2 + r^2)(d\theta^2 + \sin^2 \theta d\phi^2)$. (a) Embed the equatorial spatial slice in $\mathbb{R}^3$ via $z(r), \rho(r)$. (b) Show $\rho = R \cosh(z/R)$ solves the embedding; sketch and explain “wormhole”. (c) Geodesics; Christoffel symbols; proper time $\Delta\tau = 2r_0/V$ for radial transit. (d) Equatorial null geodesics: $u = 1/r$ obeys $u'' + u = 2R^2[(\alpha R)^2 - 1]u^3 + 2(\alpha R)^2 u$. (e) Solve perturbatively, identify b_1, b_2, b_3 , explain the absence of lensing.

(a) Embedding the slice $t, \theta = \pi/2$

Spatial 2-slice with $t = \text{const}$, $\theta = \pi/2$:

$$dS^2 = dr^2 + (R^2 + r^2) d\phi^2.$$

Embed in flat $\mathbb{R}^3$ with cylindrical coords (ρ, ϕ, z) : $dS^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2$. Match coefficients of $d\phi^2$:

$$\rho^2(r) = R^2 + r^2.$$

Match the radial sector along a curve $z(r), \rho(r)$:

$$\rho'(r)^2 + z'(r)^2 = 1.$$

Differentiate $\rho^2 = R^2 + r^2$:

$$\rho\rho' = r, \quad \rho' = \frac{r}{\sqrt{R^2 + r^2}}.$$

Insert:

$$z'(r)^2 = 1 - \frac{r^2}{R^2 + r^2} = \frac{R^2}{R^2 + r^2}.$$

$$\rho(r) = \sqrt{R^2 + r^2}, \quad z(r) = \int \frac{R dr}{\sqrt{R^2 + r^2}} = R \operatorname{arcsinh}(r/R).$$

(17)

(b) Catenoid throat

Invert $z = R \operatorname{arcsinh}(r/R)$:

$$r/R = \sinh(z/R).$$

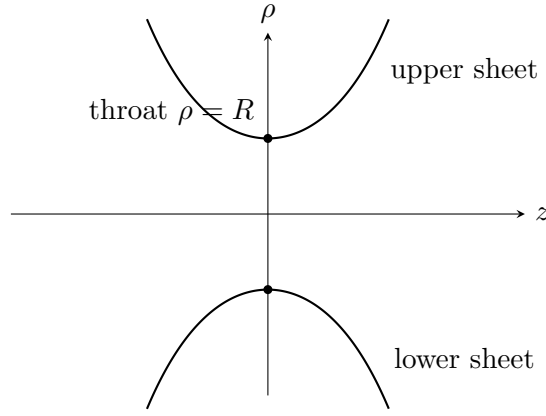
Substitute into $\rho = \sqrt{R^2 + r^2}$:

$$\rho = R\sqrt{1 + \sinh^2(z/R)} = R \cosh(z/R).$$

$\rho(z) = R \cosh(z/R).$

(18)

Sketch.



Two asymptotic flat regions ($r \rightarrow \pm\infty$) joined by a finite-radius throat at $r = 0$ ($\rho_{\min} = R$). Two universes are connected by a tube — a wormhole.

(c) Geodesics and transit time

Equatorial geodesic Lagrangian (massive, $\theta = \pi/2$):

$$2\mathcal{L} = -c^2 \dot{t}^2 + \dot{r}^2 + (R^2 + r^2) \dot{\phi}^2.$$

t, ϕ cyclic; $\theta = \pi/2$ stable. Conserved energy $\mathcal{E} = c^2 \dot{t}$ and angular momentum $\mathcal{L}_\phi = (R^2 + r^2) \dot{\phi}$. Christoffels (only non-trivial ones come from the r -dependence of $g_{\theta\theta}, g_{\phi\phi}$):

$$\Gamma_{\theta\theta}^r = -r, \quad \Gamma_{\phi\phi}^r = -r \sin^2 \theta.$$

$$\Gamma_{r\theta}^\theta = \Gamma_{r\phi}^\phi = \frac{r}{R^2 + r^2}.$$

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta, \quad \Gamma_{\theta\phi}^\phi = \cot \theta.$$

All others vanish.

Radial geodesic ($\dot{\phi} = \dot{\theta} = 0$). Normalisation $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -c^2$:

$$-c^2\dot{t}^2 + \dot{r}^2 = -c^2.$$

Inspect radial Christoffels: $\Gamma_{tt}^r = -\frac{1}{2}g^{rr}\partial_r g_{tt} = 0$ (since $g_{tt} = -c^2$ constant), $\Gamma_{rr}^r = \frac{1}{2}g^{rr}\partial_r g_{rr} = 0$ (since $g_{rr} = 1$). Radial geodesic equation:

$$\ddot{r} = 0.$$

Integrate: $\dot{r}$ constant. Initial condition $|\dot{r}| = V$ at $r = r_0$ (infalling):

$$\dot{r} = -V.$$

Proper time element:

$$d\tau = \frac{dr}{\dot{r}} = -\frac{dr}{V}.$$

Integrate from $r = r_0$ to $r = -r_0$:

$$\Delta\tau = -\frac{1}{V}\int_{r_0}^{-r_0} dr = \frac{2r_0}{V}.$$

$$\boxed{\Delta\tau = \frac{2r_0}{V}}. \quad (19)$$

(d) Null equatorial orbit equation

Photon, $\theta = \pi/2$, conserved

$$\mathcal{E} = c^2\dot{t}, \quad \mathcal{L}_\phi = (R^2 + r^2)\dot{\phi}.$$

Null condition:

$$0 = -c^2\dot{t}^2 + \dot{r}^2 + (R^2 + r^2)\dot{\phi}^2.$$

Solve for $\dot{r}^2$:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - (R^2 + r^2)\dot{\phi}^2.$$

Substitute $\dot{\phi} = \mathcal{L}_\phi/(R^2 + r^2)$:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \frac{\mathcal{L}_\phi^2}{R^2 + r^2}.$$

Define $\alpha = \mathcal{E}/(c\mathcal{L}_\phi)$ (impact parameter $1/\alpha$ in the asymptotic region):

$$\dot{r}^2 = \mathcal{L}_\phi^2 \left[\alpha^2 - \frac{1}{R^2 + r^2} \right].$$

Use $dr/d\phi = \dot{r}/\dot{\phi} = \dot{r}(R^2 + r^2)/\mathcal{L}_\phi$:

$$\left(\frac{dr}{d\phi} \right)^2 = (R^2 + r^2)^2 \left[\alpha^2 - \frac{1}{R^2 + r^2} \right] = \alpha^2(R^2 + r^2)^2 - (R^2 + r^2).$$

Substitute $u = 1/r$, $dr/d\phi = -u^{-2} du/d\phi$:

$$u^{-4} \left(\frac{du}{d\phi} \right)^2 = \alpha^2 \left(R^2 + \frac{1}{u^2} \right)^2 - \left(R^2 + \frac{1}{u^2} \right).$$

Multiply through by u^4 :

$$\left(\frac{du}{d\phi} \right)^2 = \alpha^2 (R^2 u^2 + 1)^2 - u^2 (R^2 u^2 + 1).$$

Differentiate both sides w.r.t. ϕ :

$$2 \frac{du}{d\phi} \frac{d^2u}{d\phi^2} = \alpha^2 \cdot 2(R^2u^2 + 1) \cdot 2R^2u \frac{du}{d\phi} - 2u(R^2u^2 + 1) \frac{du}{d\phi} - u^2 \cdot 2R^2u \frac{du}{d\phi}.$$

Divide both sides by $2 du/d\phi$:

$$\frac{d^2u}{d\phi^2} = 2\alpha^2 R^2u(R^2u^2 + 1) - u(R^2u^2 + 1) - R^2u^3.$$

Expand each product:

$$\frac{d^2u}{d\phi^2} = 2\alpha^2 R^4u^3 + 2\alpha^2 R^2u - R^2u^3 - u - R^2u^3.$$

Collect like terms:

$$\frac{d^2u}{d\phi^2} = 2\alpha^2 R^4u^3 + 2\alpha^2 R^2u - 2R^2u^3 - u.$$

Add u to both sides:

$$\frac{d^2u}{d\phi^2} + u = 2\alpha^2 R^4u^3 - 2R^2u^3 + 2\alpha^2 R^2u.$$

Factor $2R^2$ from the cubic terms:

$$\boxed{\frac{d^2u}{d\phi^2} + u = 2R^2[(\alpha R)^2 - 1]u^3 + 2(\alpha R)^2u.} \quad (20)$$

(e) Perturbative solution and lensing

Flat ($R = 0$) solution:

$$u_0 = \frac{\sin \phi}{b}.$$

Take $Ru \ll 1$. The orbit equation (??), treating R -terms as small, becomes

$$u'' + u = 2(\alpha R)^2u + 2R^2[(\alpha R)^2 - 1]u^3.$$

Match $\alpha = 1/b$ from the flat limit (i.e. $\alpha R = R/b \ll 1$). Drop $(\alpha R)^4$ relative to $(\alpha R)^2$:

$$u'' + u \approx 2(\alpha R)^2u - 2R^2u^3.$$

Insert ansatz $u = \sin \phi/b_1 + \sin 3\phi/b_2 + \phi \cos \phi/b_3$. Compute each term:

$$u'' = -\frac{\sin \phi}{b_1} - \frac{9 \sin 3\phi}{b_2} + \frac{-\phi \cos \phi - 2 \sin \phi}{b_3}.$$

Add u :

$$u'' + u = -\frac{8 \sin 3\phi}{b_2} - \frac{2 \sin \phi}{b_3}.$$

RHS, using $u \approx u_0 = \sin \phi/b$:

$$\begin{aligned} 2(\alpha R)^2u &= \frac{2R^2}{b^2} \cdot \frac{\sin \phi}{b} = \frac{2R^2 \sin \phi}{b^3}. \\ -2R^2u^3 &= -\frac{2R^2}{b^3} \sin^3 \phi = -\frac{2R^2}{b^3} \left[\frac{3}{4} \sin \phi - \frac{1}{4} \sin 3\phi \right]. \end{aligned}$$

Sum:

$$u'' + u = \frac{2R^2}{b^3} \sin \phi - \frac{3R^2}{2b^3} \sin \phi + \frac{R^2}{2b^3} \sin 3\phi.$$

Combine $\sin \phi$:

$$u'' + u = \frac{R^2}{2b^3} \sin \phi + \frac{R^2}{2b^3} \sin 3\phi.$$

Match to LHS coefficients:

$$\begin{aligned} -\frac{8}{b_2} &= \frac{R^2}{2b^3} \Rightarrow b_2 = -\frac{16b^3}{R^2}. \\ -\frac{2}{b_3} &= \frac{R^2}{2b^3} \Rightarrow b_3 = -\frac{4b^3}{R^2}. \end{aligned}$$

The $\sin \phi/b_1$ piece is absorbed into the homogeneous solution; matching $u \rightarrow \sin \phi/b$ as $R \rightarrow 0$ fixes

$$b_1 = b.$$

$$\boxed{b_1 = b, \quad b_2 = -\frac{16b^3}{R^2}, \quad b_3 = -\frac{4b^3}{R^2}.} \quad (21)$$

No lensing. The flat-space asymptotes are $u \rightarrow 0$ at $\phi = 0$ and $\phi = \pi$: a straight line of span π . The metric carries *no* mass parameter — there is no $1/r$ Newtonian potential and no Schwarzschild-radius analog. The $\sin 3\phi$ harmonic also vanishes at $\phi = 0$ and $\phi = \pi$, so the harmonic correction does not shift the zeros. The remaining secular $\phi \cos \phi$ piece is itself only the perturbative form of a small $\mathcal{O}(R^2/b^2)$ frequency shift, with no Newtonian mass scale entering. Physically: the bending is purely geometric (a topology effect of the throat), not gravitational — there is no monopole gravitational source, so no Newtonian-type light deflection.

Question 4 — FRW cosmology

Problem. FRW metric with curvature k . (a) Christoffel symbols. (b) Einstein equations for a perfect fluid plus Λ . (c) Age t_0 in terms of $H_0, \Omega_M, \Omega_\Lambda$; evaluate three special cases and discuss (iii). (d) Qualitative fates for $(\Omega_M > 1, \Lambda = 0)$, $(\Omega_M < 1, \Lambda = 0)$, $(\Omega_M = 1, \Lambda > 0)$. (e) Angular size of an object of size R at redshift z : smaller in $k < 0$ than $k > 0$. (f) How CMB hot/cold-spot size sets the geometry.

(a) Christoffel symbols

Non-zero metric components:

$$g_{tt} = -c^2, \quad g_{rr} = \frac{a^2}{1 - kr^2}, \quad g_{\theta\theta} = a^2 r^2, \quad g_{\phi\phi} = a^2 r^2 \sin^2 \theta.$$

Standard formula. Time-spatial:

$$\Gamma_{rr}^t = \frac{a\dot{a}}{c^2(1 - kr^2)}, \quad \Gamma_{\theta\theta}^t = \frac{a\dot{a}r^2}{c^2}, \quad \Gamma_{\phi\phi}^t = \frac{a\dot{a}r^2 \sin^2 \theta}{c^2}.$$

Spatial-time (mixed):

$$\Gamma_{tr}^r = \Gamma_{t\theta}^\theta = \Gamma_{t\phi}^\phi = \frac{\dot{a}}{a}.$$

Pure spatial:

$$\Gamma_{rr}^r = \frac{kr}{1-kr^2}, \quad \Gamma_{\theta\theta}^r = -r(1-kr^2), \quad \Gamma_{\phi\phi}^r = -r(1-kr^2)\sin^2\theta.$$

$$\Gamma_{r\theta}^\theta = \Gamma_{r\phi}^\phi = \frac{1}{r}, \quad \Gamma_{\phi\phi}^\theta = -\sin\theta\cos\theta, \quad \Gamma_{\theta\phi}^\phi = \cot\theta.$$

All others vanish. Derivation: vary the geodesic Lagrangian $2\mathcal{L} = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ for each coordinate; e.g. for t :

$$\frac{d}{d\tau}(-c^2\dot{t}) - \frac{1}{2}\partial_t g_{ij}\dot{x}^i\dot{x}^j = 0,$$

read off $\Gamma_{ij}^t = \frac{1}{2c^2}\partial_t g_{ij}$, etc.

(b) Friedmann equations

FRW Ricci tensor (under the paper's Ricci convention $R_{\alpha\beta} \equiv -R^\gamma{}_{\alpha\beta\gamma}$):

$$R_{tt} = \frac{3\ddot{a}}{a}, \quad R_{ij} = -\left[\frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2}\right]\frac{g_{ij}}{a^2}.$$

Take the trace $R = g^{\mu\nu}R_{\mu\nu}$ with $g^{tt} = -1/c^2$ and $g^{ij}g_{ij} = 3$:

$$R = -\frac{6}{c^2}\left[\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right].$$

Perfect fluid stress-energy in the comoving frame:

$$T^t{}_t = -\rho c^2, \quad T^i{}_j = p\delta_j^i.$$

Lower with metric: $T_{tt} = g_{tt}T^t{}_t = (-c^2)(-\rho c^2) = \rho c^4$, $T_{ij} = p g_{ij}$.

tt component of Einstein equation $R_{tt} - \frac{1}{2}g_{tt}R - \Lambda g_{tt} = -(8\pi G/c^4)T_{tt}$:

$$\frac{3\ddot{a}}{a} + \frac{c^2}{2}R + \Lambda c^2 = -8\pi G\rho.$$

Insert R (the $\ddot{a}/a$ terms cancel):

$$\frac{3\ddot{a}}{a} - 3\left[\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right] + \Lambda c^2 = -8\pi G\rho.$$

Simplify:

$$-\frac{3\dot{a}^2}{a^2} - \frac{3kc^2}{a^2} + \Lambda c^2 = -8\pi G\rho.$$

Multiply by $-a^2/3$ and rearrange:

$$\boxed{\dot{a}^2 = \frac{8\pi G}{3}\rho a^2 + \frac{1}{3}\Lambda c^2 a^2 - kc^2.} \quad (22)$$

Spatial-trace Einstein equation, contract with g^{ij} (using $g^{ij}g_{ij} = 3$, $g^{ij}T_{ij} = 3p$):

$$-3\left[\frac{a\ddot{a} + 2\dot{a}^2 + 2kc^2}{c^2 a^2}\right] - \frac{3R}{2} - 3\Lambda = -\frac{24\pi Gp}{c^4}.$$

Insert R and simplify:

$$\frac{2\ddot{a}}{ac^2} + \frac{\dot{a}^2}{a^2 c^2} + \frac{k}{a^2} - \Lambda = -\frac{8\pi Gp}{c^4}.$$

Use (??) to eliminate $\dot{a}^2/a^2$:

$$\frac{2\ddot{a}}{ac^2} + \frac{8\pi G\rho}{3c^2} + \frac{\Lambda}{3} - \frac{k}{a^2} + \frac{k}{a^2} - \Lambda = -\frac{8\pi Gp}{c^4}.$$

Cancel k/a^2 , collect Λ :

$$\frac{2\ddot{a}}{ac^2} + \frac{8\pi G\rho}{3c^2} - \frac{2\Lambda}{3} = -\frac{8\pi Gp}{c^4}.$$

Multiply by $ac^2/2$ and rearrange:

$$\boxed{\ddot{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right)a + \frac{1}{3}\Lambda c^2 a.} \quad (23)$$

(Identifying $a \leftrightarrow R$ in the paper.)

(c) Age of the universe

Define $H = \dot{a}/a$, $\rho_c = 3H_0^2/(8\pi G)$, $\Omega_M = \rho_M/\rho_c|_0$, $\Omega_\Lambda = \Lambda c^2/(3H_0^2)$, $\Omega_k = -kc^2/(a_0 H_0)^2$. Matter scales as $\rho_M \propto a^{-3}$, so $\rho_M(a) = \rho_{M,0}(a_0/a)^3$. Insert into (??):

$$H^2 = H_0^2 [\Omega_M (a_0/a)^3 + \Omega_\Lambda + \Omega_k (a_0/a)^2].$$

Use redshift $1+z = a_0/a$, $da/a = -dz/(1+z)$, $H = da/(a dt)$:

$$dt = -\frac{dz}{(1+z)H(z)}.$$

Integrate from $z = \infty$ (Big Bang) to $z = 0$ (today):

$$\boxed{t_0 = \frac{1}{H_0} \int_0^\infty \frac{dz}{(1+z)\sqrt{\Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda}}.} \quad (24)$$

(i) $\Omega_M = 1, \Omega_\Lambda = 0$ (*Einstein-de Sitter*, $\Omega_k = 0$). Integrand simplifies to $(1+z)^{-5/2}$:

$$t_0 = \frac{1}{H_0} \int_0^\infty (1+z)^{-5/2} dz.$$

Antiderivative:

$$t_0 = \frac{1}{H_0} \left[-\frac{2}{3}(1+z)^{-3/2} \right]_0^\infty = \frac{2}{3H_0}.$$

$$\boxed{t_0 = \frac{2}{3H_0}.$$

(ii) $\Omega_M = 0, \Omega_\Lambda = 0$ (*Milne*, $\Omega_k = 1$). Integrand $(1+z)^{-2}$:

$$t_0 = \frac{1}{H_0} \int_0^\infty (1+z)^{-2} dz = \frac{1}{H_0}.$$

$$\boxed{t_0 = \frac{1}{H_0}.$$

(iii) $\Omega_M = 0, \Omega_\Lambda = 1$ (*de Sitter*, $\Omega_k = 0$). Integrand $(1+z)^{-1}$:

$$t_0 = \frac{1}{H_0} \int_0^\infty \frac{dz}{1+z}.$$

Diverges:

$$\boxed{t_0 = \infty.}$$

Discussion. Pure de Sitter expansion has no Big Bang: $a(t) = a_0 e^{H_0 t}$ for all $t \in (-\infty, \infty)$. There is no past singularity — only an exponentially expanding space sourced entirely by vacuum energy. The integral diverges logarithmically because, going back in time, $a \rightarrow 0$ is never reached at finite t .

(d) Fates

(i) $\Omega_M > 1, \Lambda = 0$: closed, $k > 0$. Friedmann (??) has $\dot{a}^2$ vanishing at finite $a_{\max}$; thereafter $\ddot{a} < 0$ from (??), so the universe recollapses — Big Crunch.

(ii) $\Omega_M < 1, \Lambda = 0$: open, $k < 0$. $\dot{a}^2 \rightarrow -kc^2 > 0$ as $a \rightarrow \infty$; $\ddot{a} < 0$ but never zero. Eternal expansion, asymptotically coasting at $\dot{a} \rightarrow c\sqrt{|k|}$.

(iii) $\Omega_M = 1, \Lambda > 0$: Λ dominates at late times. Initially matter dominated ($a \propto t^{2/3}$); transitions to de Sitter $a \propto e^{Ht}$ when $\rho_M \sim \rho_\Lambda$. Eternal accelerating expansion.

(e) Angular size in spatially curved FRW

Comoving radial distance from null geodesic ($ds = 0$, radial):

$$D_c = \int_0^z \frac{c dz'}{H(z')} \quad (\text{same in both universes by assumption}).$$

The spatial 3-slice has $d\Sigma^2 = a^2[dr^2/(1 - kr^2) + r^2 d\Omega^2]$. Relate the FRW radial coordinate r to comoving distance $\chi = D_c/a_0$ via the radial proper-distance integral:

$$\chi = \int_0^r \frac{dr'}{\sqrt{1 - kr'^2}}.$$

For $k > 0$, apply hint $\int dx/\sqrt{1 - x^2} = \arcsin x$ with $x = \sqrt{k}r'$:

$$\sqrt{k}\chi = \arcsin(\sqrt{k}r).$$

Invert:

$$r = \frac{1}{\sqrt{k}} \sin(\sqrt{k}\chi).$$

For $k < 0$, apply hint $\int dx/\sqrt{1 + x^2} = \sinh^{-1} x$ with $x = \sqrt{|k|}r'$:

$$\sqrt{|k|}\chi = (\sqrt{|k|}r).$$

Invert:

$$r = \frac{1}{\sqrt{|k|}} \sinh(\sqrt{|k|}\chi).$$

Transverse physical size at the source is $R = a(t_e)r\theta$, giving angular size $\theta = R/D_A$ with $D_A = a_0r/(1 + z)$. Define transverse comoving distance $D_M = a_0r$. Using $\Omega_k = -kc^2/(a_0H_0)^2$, so $\sqrt{|k|}\chi = |\Omega_k|^{1/2}H_0D_c/c$ and $a_0/\sqrt{|k|} = (c/H_0)|\Omega_k|^{-1/2}$:

$$D_M = \begin{cases} (c/H_0)|\Omega_k|^{-1/2} \sin[|\Omega_k|^{1/2}H_0D_c/c], & k > 0, \\ D_c, & k = 0, \\ (c/H_0)|\Omega_k|^{-1/2} \sinh[|\Omega_k|^{1/2}H_0D_c/c], & k < 0. \end{cases}$$

Angular size:

$$\theta = \frac{R}{D_A} = \frac{R(1 + z)}{D_M}.$$

Expand for small $x = |\Omega_k|^{1/2}H_0D_c/c \ll 1$:

$$\sin x \approx x - \frac{x^3}{6}, \quad \sinh x \approx x + \frac{x^3}{6}.$$

Hence to first order in curvature:

$$D_M^{(k>0)} < D_c < D_M^{(k<0)}.$$

Invert via $\theta \propto 1/D_M$:

$$\theta^{(k<0)} < \theta^{(k=0)} < \theta^{(k>0)}.$$

$$\boxed{\theta(k < 0) < \theta(k > 0) \text{ at fixed } D_c.} \quad (25)$$

Geometric interpretation: positive curvature focuses light (acts as a converging lens enlarging the image), negative curvature defocuses.

(f) Geometry from CMB hot-spot size

The sound horizon at recombination, $r_s \approx 147 \text{ Mpc}$, sets the physical size R of the largest acoustic feature, frozen at $z \simeq 1080$. Its angular size on the sky, θ_* , is measured directly from the position of the first peak in the CMB temperature angular power spectrum (or visually as the typical hot/cold-spot scale, $\sim 1^\circ$). Compute $D_A(z = 1080)$ from $\theta_* = R/D_A$. Compare this to the D_A predicted from (??)-style integrals for $k = 0$, $k > 0$, $k < 0$ at fixed Ω_M, Ω_Λ :

- $\theta_* < 1^\circ$: spots smaller than flat prediction $\Rightarrow k < 0$ (open).
- $\theta_* \approx 1^\circ$: $\Rightarrow k \approx 0$ (flat). Observed.
- $\theta_* > 1^\circ$: $\Rightarrow k > 0$ (closed).

Planck measurement gives $\theta_* \approx 1^\circ$, consistent with $\Omega_k = 0$ to $\sim 0.4\%$.